1 - Synthetic data generation

October 8, 2024

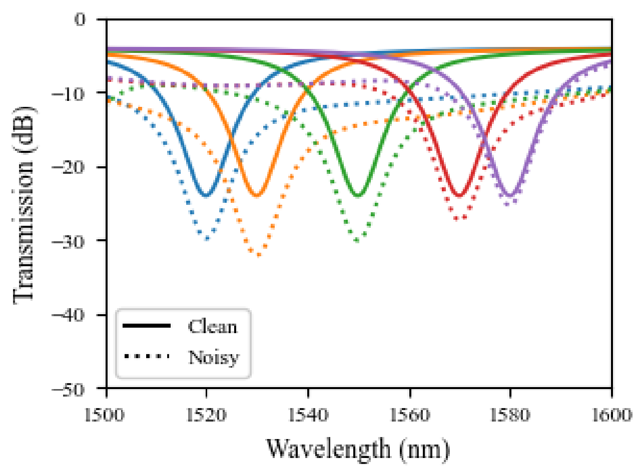
1 Synthetic spectra visualization

Some examples of synthetic LPFG spectra

1.1 Varying the resonant wavelength

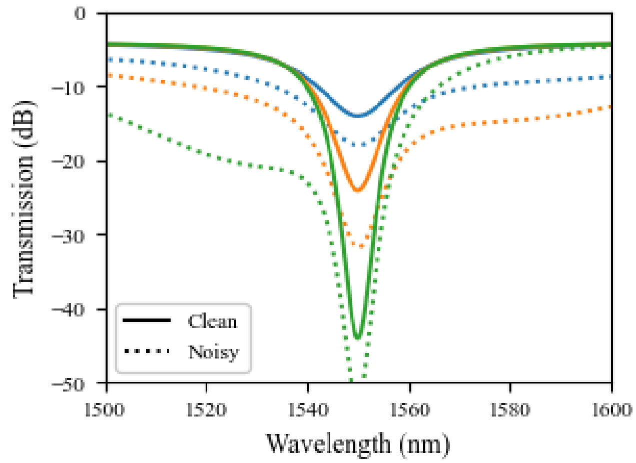
Below one can see some LPFG spectra with different resonant wavelength.

Both ideal/clean spectra and noisy spectra. Noise spectra were generate by adding extra attenuation bands with random parameters.



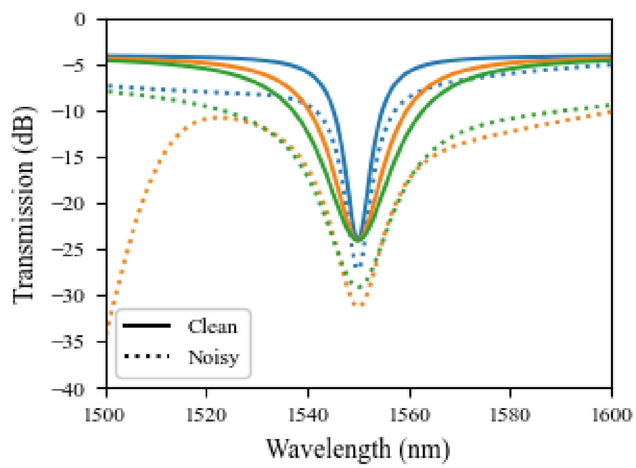
1.2 Varying the coupling efficiency

A similar demonstration, however, with change in the resonant dip depth.



1.3 Varying the FWHM

Now varying the resonant dip's width



2 Data generation

Note that the model uses the FBG reflection. So, for training, the whole spectrum is not needed.

For time and memory efficiency, the synthetic LPFG is only simulated at the FBG position.

Below an example of data generation using the same parameters as the paper.

```
100%|      | 100/100 [00:00<00:00, 2034.43it/s]
```

```
Dataset stored as a dictionary with keys:
dict_keys(['input_strength', 'input_strength_clean', 'wl_bragg', 'target'])
```

2.1 Show some random synthetic LPFG filtered FBG peaks